

Dataset Files

Dataset Format

We provide all datasets in two formats: text files and binary files (rosbag). While their content is identical, some of them are better suited for particular applications. The binary [rosbag](#) files are intended for users familiar with the [Robot Operating System \(ROS\)](#) and for applications that are intended to be executed on a real system.

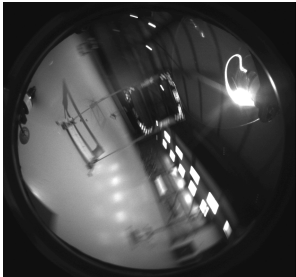
Binary Files (rosbag)

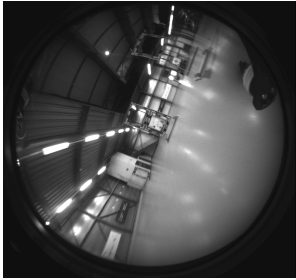
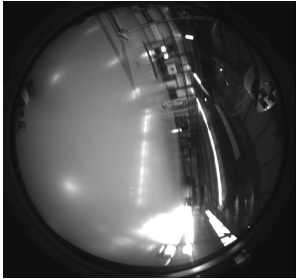
The rosbag files contain images and IMU measurements using the standard [sensor_msgs/Image](#) and [sensor_msgs/Imu](#) message types, respectively. The events are provided as [dvs_msgs/EventArray](#) message types, and the ground truth is provided as [geometry_msgs/PoseStamped](#) messages. The Events/IMU/GT bag files also contain the image frames from the mDAVIS as [sensor_msgs/Image](#) messages.

Sequences

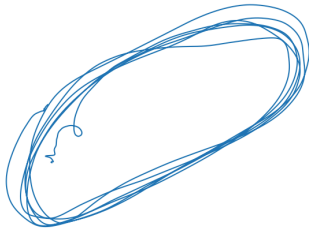
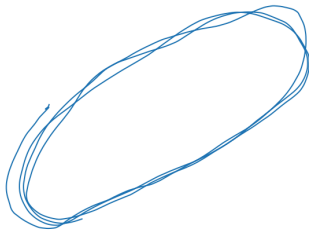
Note that some ground truth files are withheld for benchmarks and competitions.

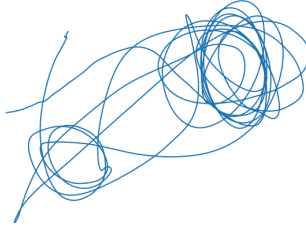
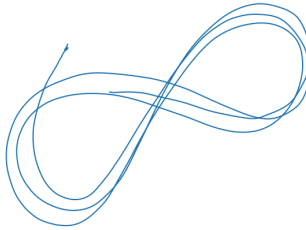
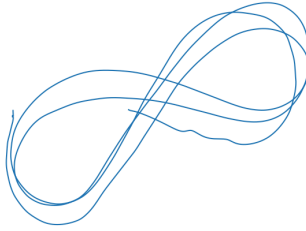
SplitS Track

#	Duration (s)	Length (m)	vmax (m/s)	Realsense T265		Ground Truth Public*
				Rosbag	ZIP	
1	28.30	246.28	26.79	587.0MB	287.2MB	no
						Very Hard

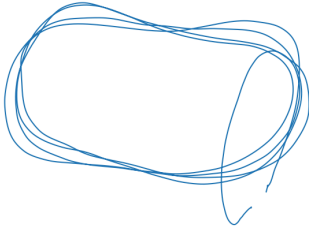
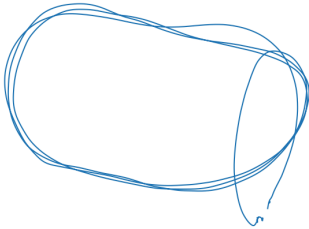
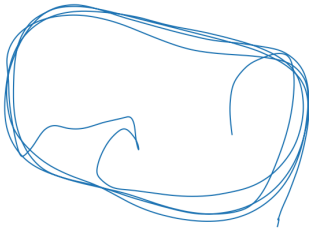
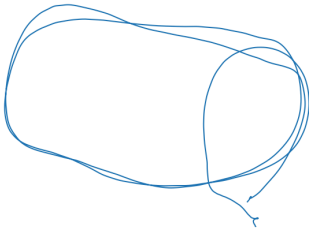
2	30.17	246.38	25.90	596.7MB	271.7MB	no
						Very Hard
3	27.26	249.05	20.03	564.4MB	254.3MB	no
						Very Hard

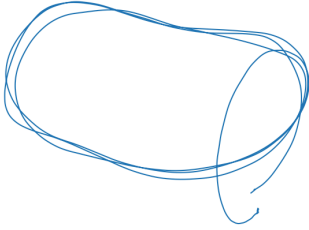
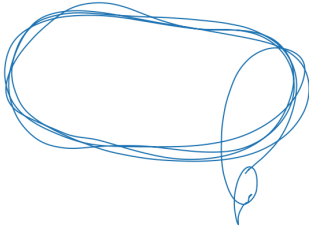
Indoor forward facing

#	Duration (s)	Length (m)	vmax (m/s)	DAVIS		Snapdragon		Leica	Groun Truth Public
				Rosbag	ZIP	Rosbag	ZIP		
3	54.63	287.12	9.5	853.7MB	413.9MB	1.6GB	842.8MB	76kB	yes
								Easy	
5	50	156.47	4.87	1.3GB	655.3MB	2.6GB	1.3GB	142kB	yes
								Mediu	
6	32.93	223.27	12.52	728.1MB	365.1MB	1.2GB	605.3MB	58kB	yes

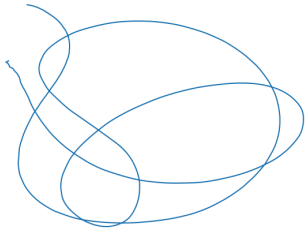
									Mediu
7	73.2	333.59	12.78	1.1GB	559.5MB	2GB	964.1MB	95kB	yes
									Hard
8	132.53	259.16	5.26	1.4GB	726.9MB	2.8GB	1.4GB	–	no
9	34.04	157.07	11.42	710.5MB	365.6MB	1.3GB	627MB	68kB	yes
									Easy
10	33.43	149.36	9.49	643.7MB	327.9MB	1.3GB	678.7MB	69kB	yes
									Easy
11	24.02	85.68	10.32	973.2MB	467.5MB	1.4GB	731.3MB	–	no
12	31.98	124.07	15.28	565.1MB	268.4MB	1GB	567.8MB	–	no

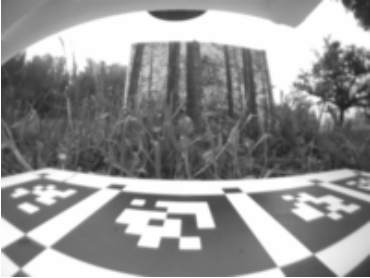
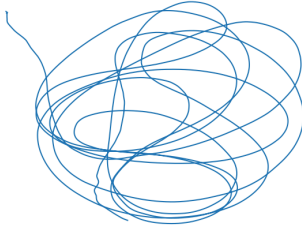
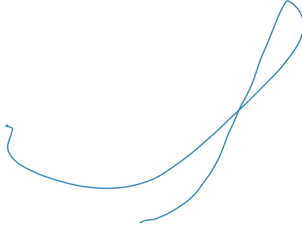
Indoor 45 degree downward facing

#	Duration (s)	Length (m)	vmax (m/s)	DAVIS		Snapdragon		Leica	Groun Truth Public
				Rosbag	ZIP	Rosbag	ZIP		
1	73.99	150.71	4.36	1.3GB	635.8MB	1.6GB	897.3MB	–	no
2	55.77	218.9	6.97	1.3GB	602.5MB	1.2GB	643.5MB	83kB	yes
								Easy	
3	57.82	119.82	3.53	870.6MB	437.8MB	1.3GB	683.1MB	–	no
4	47.36	168.06	6.55	1.1GB	496.3MB	1.1GB	607.8MB	71kB	yes
								Easy	
9	40	215.58	11.23	808.1MB	370.8MB	1.2GB	664MB	62kB	yes
								Mediu	
11	22.96	125.21	11.74	949.6MB	440.1MB	893.3MB	480.8MB	–	no
12	51.25	124.56	4.33	563.7MB	268.8MB	1.3GB	648.6MB	72kB	yes
								Easy	

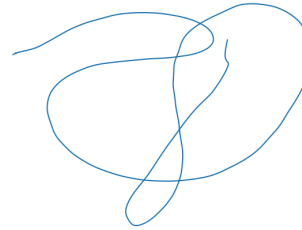
13	42.49	166.62	7.92	527.9MB	254.7MB	1.1GB	534.9MB	62kB	yes
									Medium
14	43.66	220.4	9.54	625.6MB	301MB	1.2GB	543.8MB	60kB	yes
									Hard
16	15.49	58.72	7.69	336MB	166.1MB	805.3MB	448MB	–	no

Outdoor forward facing

#	Duration (s)	Length (m)	vmax (m/s)	DAVIS		Snapdragon		Leica	Ground Truth Public
				Rosbag	ZIP	Rosbag	ZIP		
1	49.63	258.23	8.55	687.2MB	358.6MB	1.5GB	664.8MB	123kB	yes
									Easy
2	36.9	220.88	10.13	763.4MB	410.3MB	249.8MB	710.4MB	–	no
3	92.84	735.51	14.04	1.1GB	637.2MB	2.2GB	1.1GB	117kB	yes

									Mediu
5	22.21	189.63	20.73	657.1MB	342.7MB	1.5GB	729.6MB	72kB	yes
									Hard
6	34.83	338.2	19.42	786.5MB	427MB	1.3GB	677.6MB	–	no
9	43.15	314.41	10.68	1.1GB	577.4MB	1.5GB	742.1MB	–	no
10	59.6	455.63	12.58	1.2GB	656MB	1.8GB	955.6MB	–	no

Outdoor 45 degree downward facing

#	Duration (s)	Length (m)	vmax (m/s)	DAVIS		Snapdragon		Leica	Ground Truth Public'
				Rosbag	ZIP	Rosbag	ZIP		
1	24.49	165.53	15.62	1.3GB	650.4MB	1.4GB	723.3MB	46kB	yes
									Mediu
2	26.19	143.13	10.68	1.1GB	588.1MB	1.2GB	633.5MB	–	no

Calibration

We provide the calibration parameters for the camera intrinsics and camera-IMU extrinsics in YAML format, as well as the raw calibration sequences used to produce

those with the [Kalibr toolbox](#).

For sequences	DAVIS			Snapdragon		
	Calib. sequences		Calibration	Calib. sequences		Calibration
	Camera	IMU		Camera	IMU	
Indoor forward	134.1MB	138.6MB	2.2MB	1.2GB	1GB	4.1MB
Indoor 45 down	206.8MB	112.8MB	2.2MB	1.3GB	792.3MB	2.2MB
Outdoor forward	213.1MB	227.6MB	4.4MB	919.1MB	892.9MB	5.3MB
Outdoor 45 down	214.7MB	130.1MB	2.3MB	1.4GB	578.4MB	3.5MB

For sequences	Realsense T265	
	Calib. sequence	Calibration
SplitS	1.1GB	3.3MB

License

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